DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 3: Assignment

Mean-Variance Analysis · The Efficient Frontier · Two-Fund Separation

Instructions

- This assignment covers Lectures 7–9 and the Week 3 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script or notebook cell; submit your code alongside your written answers.
- Submit a single PDF (typeset or clearly scanned handwriting) plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (20 marks)

A1. **(4 marks)** Explain, in 2–3 sentences, why portfolio variance depends on the *covariances* between assets, not merely on their individual variances.

A2. **(4 marks)** Without any calculation, sketch (describe in words) how the two-asset opportunity set changes shape as correlation moves from $\rho = +1$ to $\rho = 0$ to $\rho = -1$.

A3. **(4 marks)** True or false, with justification: "The lower branch of the Markowitz bullet is feasible but never optimal."

A4. **(4 marks)** Explain why the minimum variance portfolio can be computed without any estimate of expected returns, while every other point on the efficient frontier requires one.

A5. **(4 marks)** State the Two-Fund Separation Theorem in your own words, in no more than three sentences.

Part B – Two-Asset and Matrix Portfolio Calculations (25 marks)

B1. **(6 marks)** Asset A: $\mu_A = 12\%, \sigma_A = 18\%$. Asset B: $\mu_B = 20\%, \sigma_B = 35\%$. Correlation $\rho = 0.2$. For weights $w_A = 0.7, w_B = 0.3$, compute μ_p and σ_p^2 (and hence σ_p).

B2. **(6 marks)** Write the covariance matrix Σ for the same two assets in B1 (using $\Sigma_{12} = \rho\sigma_1\sigma_2$), and compute $\sigma_p^2 = w^\top \Sigma w$ for the same weights directly from the matrix. Confirm it matches your answer to B1.

B3. **(7 marks)** Three assets have the covariance matrix

$$\Sigma = \begin{pmatrix} 225 & 45 & 20 \\ 45 & 625 & 90 \\ 20 & 90 & 400 \end{pmatrix} \ (\%^2).$$

For weights $w = (0.4, 0.35, 0.25)^\top$, compute σ_p^2 and σ_p, showing every cross-term in the expansion of $w^\top \Sigma w$.

B4. **(6 marks)** Explain algebraically why a candidate "portfolio" with weights that do not sum to 1 (e.g., $w = (0.5, 0.5, 0.5)^\top$) is not a valid portfolio in the sense used in this course, connecting your answer to the constraint $\mathbf{1}^\top w = 1$ in the Markowitz problem.

Part C – The Efficient Frontier (20 marks)

C1. **(6 marks)** Four portfolios have been computed: W (σ=10%, μ=11%), X (σ=10%, μ=9%), Y (σ=14%, μ=13%), Z (σ=18%, μ=12%). Which of these portfolios could plausibly lie on the efficient frontier, and which are definitely dominated? Justify each answer by naming the portfolio that dominates it, where applicable.

C2. **(5 marks)** Using the "more assets can only help" argument from Lecture 8, explain why a fund holding 500 stocks generally offers a more favourable efficient frontier than one holding only 5 – even before considering which specific 500 or 5 stocks are chosen.

C3. **(5 marks)** Write out (do not solve) the full quadratic program for finding the minimum-variance portfolio with target return exactly 15%, for a generic 4-asset covariance matrix Σ and mean vector μ. State the objective and all constraints explicitly.

C4. **(4 marks)** Explain, referencing the structure of $w^\top \Sigma w$, why the boundary of the feasible set traces out a hyperbola-like curve in (σ_p, μ_p) space rather than, say, a straight line.

Part D – Two-Fund Separation and the CML (20 marks)

D1. **(4 marks)** $r_f = 5\%$. Tangency portfolio T has $\mu_T = 16\%, \sigma_T = 22\%$. Compute T's Sharpe ratio.

D2. **(6 marks)** An investor wants a combined portfolio with $\sigma_C = 11\%$, using the r_f and T from D1. Find the required weight y in T, and the resulting μ_C.

D3. **(6 marks)** A more aggressive investor instead wants $\mu_C = 25\%$, using the same r_f and T. Find y (note it may exceed 1) and the resulting σ_C. Explain what $y > 1$ means in practice.

D4. **(4 marks)** Explain, referencing Week 2's utility framework, why two investors with very different CRRA risk-aversion parameters γ should still hold their risky assets in *exactly* the same relative proportions as each other (i.e., both holding only T as their risky fund).

Part E – Computational (10 marks)

E1. **(10 marks) [Python]** Using your Lab code (or a fresh, self-contained script) and the bundled `sample_nifty_prices.csv` data, compute the tangency portfolio's composition and Sharpe ratio for four different risk-free rates: $r_f \in \{0\%, 3\%, 6\%, 9\%\}$. Present your results in a small table (weights and Sharpe ratio for each r_f), and comment in 2–3 sentences on how the tangency portfolio's composition and Sharpe ratio change as r_f increases.

Part F – Integrative Question (5 marks)

F1. **(5 marks)** In a short paragraph (100–150 words), explain why a risk-averse investor with CRRA utility (Week 2) would rationally combine the tangency portfolio (Week 3) with government bonds (Week 1's fixed, near-risk-free claim) – rather than either investing entirely

in the tangency portfolio, or hand-picking individual stocks directly. Your answer should reference the Two-Fund Separation Theorem and the role bonds play as a practical proxy for the risk-free asset.

End of Assignment.